

PORTFOLIO SAMPLE

Scientific LaTeX Technical Writing Sample

Quality-Aware Federated Regression at the Edge

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Scientific writing • LaTeX/Overleaf • Mathematical typesetting • Research communication

Purpose. This document is an original, non-confidential portfolio demonstration. It is designed to show scientific writing and LaTeX skills—including equations, notation, cross-references, algorithms, tables, figures, citations, and bibliography management—rather than to report a new empirical study.

1. Technical Context

Federated learning (FL) enables multiple clients to optimize a shared model without pooling their raw data at a central server. In the canonical formulation, each client performs local optimization and communicates model updates for server-side aggregation (McMahan et al., 2017). This paradigm is useful when data are distributed across devices, institutions, or sensing locations and when communication or privacy constraints make centralized training undesirable (Kairouz et al., 2021).

Edge deployments introduce an additional challenge: not all client observations are equally informative. Measurements close to known sensor limits, temporary communication failures, or sparsely sampled periods may be less reliable than well-supported observations. A quality-aware objective can therefore be used to moderate the contribution of uncertain samples while retaining the decentralized training structure.

This short example develops a compact regression formulation. The technical goal is intentionally simple: predict a continuous target $y \in \mathbb{R}$ from a feature vector $\mathbf{x} \in \mathbb{R}^d$ while accounting for per-sample confidence. The emphasis is on clear scientific exposition and reproducible mathematical notation.

1.1 Problem Definition and Notation

Assume K edge clients. Client k owns a local dataset

$$\mathcal{D}_k = \{(\mathbf{x}_{k,i}, y_{k,i}, q_{k,i})\}_{i=1}^{n_k}, \quad (1)$$

where $q_{k,i} \in [0, 1]$ is a quality score and n_k is the number of local samples. Let $f_{\boldsymbol{\theta}}(\mathbf{x})$ denote a regression model parameterized by $\boldsymbol{\theta}$.

For a single sample, the weighted squared-error loss is

$$\ell_{k,i}(\boldsymbol{\theta}) = q_{k,i} (y_{k,i} - f_{\boldsymbol{\theta}}(\mathbf{x}_{k,i}))^2. \quad (2)$$

The normalized local objective becomes

$$\mathcal{L}_k(\boldsymbol{\theta}) = \frac{1}{\sum_{i=1}^{n_k} q_{k,i} + \varepsilon} \sum_{i=1}^{n_k} \ell_{k,i}(\boldsymbol{\theta}), \quad (3)$$

where $\varepsilon > 0$ avoids division by zero. The global objective is

$$\min_{\boldsymbol{\theta}} \quad \mathcal{L}(\boldsymbol{\theta}) = \sum_{k=1}^K \alpha_k \mathcal{L}_k(\boldsymbol{\theta}), \quad (4)$$

$$\alpha_k = \frac{n_k}{\sum_{j=1}^K n_j}. \quad (5)$$

Equations (3)–(5) separate two weighting mechanisms: $q_{k,i}$ controls within-client sample influence, whereas α_k controls the client contribution during aggregation.

2. System Workflow

The workflow in Fig. 1 separates sensing, local optimization, and server aggregation. Raw observations remain at the client. Only model parameters or parameter updates are transmitted.

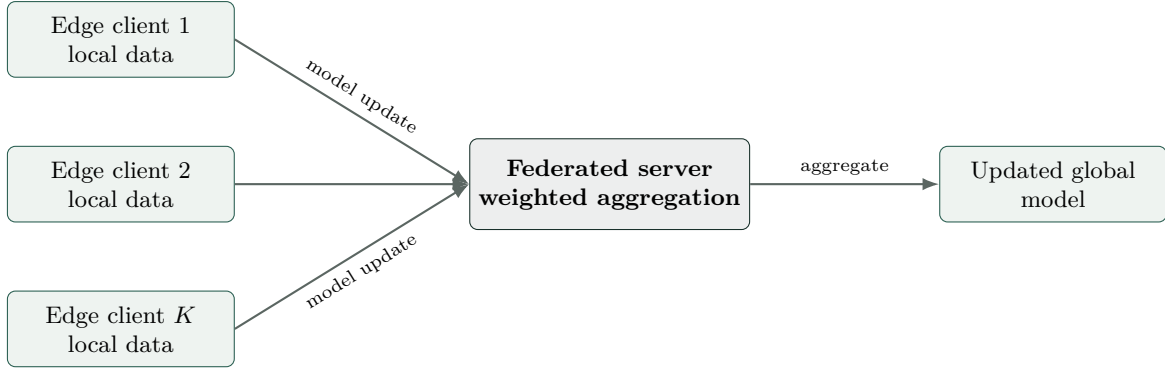


Figure 1: Illustrative federated regression workflow. Client data remain local; only model information is exchanged.

At round t , the server broadcasts $\theta^{(t)}$. Each selected client performs E local epochs and returns updated parameters $\theta_k^{(t+1)}$. A FedAvg-style aggregation step (McMahan et al., 2017) is then

$$\theta^{(t+1)} = \sum_{k \in \mathcal{S}_t} \frac{n_k}{\sum_{j \in \mathcal{S}_t} n_j} \theta_k^{(t+1)}, \quad (6)$$

where $\mathcal{S}_t$ denotes the clients participating in round t .

2.1 Algorithmic Description

Algorithm 1 gives a concise training procedure. The pseudocode makes the data flow explicit and distinguishes local sample weighting from server-side client aggregation.

Algorithm 1 Quality-Aware Federated Regression

Require: Clients $\{1, \dots, K\}$, rounds T , local epochs E , learning rate η

- 1: Initialize global parameters $\theta^{(0)}$
 - 2: **for** $t = 0, 1, \dots, T - 1$ **do**
 - 3: Server selects participating clients $\mathcal{S}_t$
 - 4: **for all** $k \in \mathcal{S}_t$ **in parallel do**
 - 5: Set $\theta_k \leftarrow \theta^{(t)}$
 - 6: **for** $e = 1, \dots, E$ **do**
 - 7: Compute $\nabla \mathcal{L}_k(\theta_k)$ using Eq. (3)
 - 8: $\theta_k \leftarrow \theta_k - \eta \nabla \mathcal{L}_k(\theta_k)$
 - 9: **end for**
 - 10: Return θ_k and n_k to server
 - 11: **end for**
 - 12: Aggregate client models using Eq. (6)
 - 13: **end for**
 - 14: **return** final global model $\theta^{(T)}$
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3. Evaluation and Reporting

A scientific report should distinguish predictive performance from deployment cost. For regression, root mean squared error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2) are common complementary metrics. RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}. \quad (7)$$

Communication can be summarized using transmitted bytes per round, whereas edge feasibility can include peak memory, local training time, and model size.

The example in Table 1 demonstrates a reporting convention rather than a research result. In a real study, numerical comparisons should be accompanied by uncertainty estimates, clearly defined train/validation/test

Table 1: Illustrative reporting structure for a federated regression experiment. Values are synthetic and included only to demonstrate scientific table formatting.

Method	RMSE	MAE	R^2	Comm. (MB)
Centralized reference	0.082	0.061	0.88	–
FedAvg	0.091	0.068	0.85	14.2
Quality-aware FL	0.086	0.064	0.87	14.2
Local only	0.108	0.079	0.78	0.0

partitions, and a statement of whether hyperparameters were tuned using validation data only. When multiple clients are evaluated, reporting both aggregate metrics and the distribution across clients helps expose performance heterogeneity.

3.1 Reproducibility Checklist

A compact reproducibility section can make technical work easier to audit. At minimum, a manuscript should specify:

- the data partitioning strategy and any exclusion criteria;
- preprocessing, normalization, and missing-data handling;
- model architecture, optimizer, learning rate, batch size, and random seeds;
- FL participation rate, number of rounds, and local epochs;
- metric definitions and the unit of aggregation; and
- software versions, hardware, and availability of code or configuration files.

These details are especially important in FL because client partitioning and local training choices can materially alter the optimization problem (Li et al., 2020).

4. Technical Writing Notes

This sample intentionally demonstrates several document-production capabilities in one short artifact:

- consistent mathematical notation across Eqs. (1), (5) and (6);
- automatic numbering and cross-referencing for equations, figures, tables, sections, and algorithms;
- a vector workflow diagram created directly in TikZ;
- a professional table using `booktabs`;
- structured pseudocode using `algorithm` and `algpseudocode`;
- bibliography management using BibTeX-compatible citations; and
- concise technical prose that distinguishes assumptions, definitions, procedures, and evidence.

The same workflow can be adapted to publisher-specific templates such as IEEE, Springer, or Elsevier while preserving consistent notation, references, figures, and document structure.

5. Conclusion

This portfolio document presents a compact example of scientific LaTeX preparation for a machine-learning topic. The emphasis is not on proposing a novel algorithm, but on demonstrating how technical content can be organized into a coherent, publication-style document with traceable definitions, equations, algorithms, figures, tables, and references.

Document Production Details

The public source package accompanying this PDF contains the complete `.tex` file and a separate `references.bib` database. The PDF is intentionally self-contained for straightforward compilation on a standard TeX installation; the bibliography database is provided to demonstrate a reusable BibTeX workflow for publisher templates. Core packages used in this sample include `amsmath`, `cleveref`, `booktabs`, `algorithmicx`, and `TikZ`.

A minimal reproducible build consists of two LaTeX passes:

```
pdflatex scientific-latex-sample.tex → pdflatex scientific-latex-sample.tex
```

References

- H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proc. 20th Int. Conf. Artificial Intelligence and Statistics*, 2017, pp. 1273–1282.
- P. Kairouz, H. B. McMahan, B. Avent, et al., “Advances and open problems in federated learning,” *Foundations and Trends in Machine Learning*, vol. 14, nos. 1–2, pp. 1–210, 2021.
- T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, “Federated optimization in heterogeneous networks,” *Proceedings of Machine Learning and Systems*, vol. 2, pp. 429–450, 2020.